Introduction to Buoyant Plume Theory (BPT)

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This resource was created to introduce the physical theory behind some of the buoyant plume models employed by oceanographers to describe subglacial freshwater plumes. The models' popularity is owed in part to their relative simplicity, and yet (as I learned all too well during the research for this section), it is hard to find a recent and comprehensive document that outlines exactly what they do and why. When using a model it is important to know what is going on behind the curtain, so this resource was written partly to serve as a stand-alone introduction to the subject for future students. (This article appeared originally as a chapter in a 2026 undergraduate honors thesis by Natalie McGee, titled "Modification of fjord waters by subglacial freshwater plumes in Upernavik Fjord, Greenland").

Code Resources

An interactive jupyter notebook was built to accompany this written introduction and provides simple runnable examples of the models: `https://github.com/nmcgee108/Intro_to_plume_models`. The Python scripts for the full and analytical models, adapted by Natalie McGee from existing Matlab scripts by Donald Slater (to accompany his 2022 work [1]) and Aurora Roth, are available in this repository as well.

A buoyant plume—the phenomenon caused by rising gas or liquid that is less dense than its surroundings—is a type of fluid motion that is observed widely in nature and everyday life. For instance, rising smoke from a chimney, steam from a kettle, or hot water rising from hydrothermal vents in the ocean are all examples of this behavior. In these cases, the buoyancy forcing comes from high temperature (hot air and water are less dense than their surroundings), but there are several variables that can impact the density of a fluid. In the ocean, notably, we observe plumes of freshwater rising through salty ambient water in much the same manner as the thermal examples above. Buoyant plume theory (BPT) proposes that all such examples can be modeled by the same mathematical framework: that is, that buoyancy forcing is the main driver of the plume's evolution, causing it to grow and change by entraining ambient fluid.

The plumes described by BPT begin with a source: typically either a point or a line where the buoyant fluid originates. In the context of Greenlandic fjords, we think of the source as the opening at the base of a glacier where buoyant subglacial discharge (SGD) is released into the fjord [2]. BPT assumes that the rising fluid is turbulent from the source, and turbulent mixing as it rises causes the buoyant plume to incorporate ambient fluid from its surroundings, a process called entrainment [3]. Entrainment of surrounding water causes the plume's width to grow and its properties (such as temperature, salinity, etc.) to become diluted towards the ambient properties as a function of height from the source. In the context of subglacial plumes, the salinity and temperature of the plume increase as the relatively warm, salty fjord water is entrained. This increases the density of the plume according to its equation of state until it is no longer buoyant enough to keep rising; the depth at which the density of the plume is approximately equal to the density of the ambient fjord water is called the plume's neutral buoyancy depth. Some plumes make it to the surface of the fjord before reaching neutral buoyancy, though many do not. In theory, a rising plume will "overshoot" its neutral buoyancy depth because of its upward momentum and will then sink back downwards towards neutral buoyancy. A plume that has leveled out below the surface will be exported away from the glacier front at its neutral buoyancy depth as a buoyancy-driven current [4].

BPT models focus on understanding the properties of a plume as it rises: far enough away from the source that an idealized geometry can apply, and before it reaches neutral buoyancy [2], [5]. Plume models also do not describe the properties down-fjord of the plume, where oceanographic measurements can be made. Their usefulness comes from the fact that they can give an idea of how variables such as SGD volume and ocean stratification influence plume properties such as temperature, salinity, and velocity. It is important to note that plume models are never "right" — no model, short of replicating the path of every particle, can recreate such complex dynamics.

Within the realm of buoyant plume theory models, there are a variety of constraints that can be applied to learn more about different cases. For our purposes in the context of oceanography, we are most interested in plumes rising from a maintained buoyancy source (the freshwater supplied at depth from SGD) and emerging from either a point or a line source, depending on the geometry of the SGD outlet. Additional assumptions are described in the following sections.

In this chapter, I will provide a brief survey of the development of BPT since the mid-20th century, following its evolution across a series of influential papers. I will describe the key assumptions made along the way, and will recreate the mathematical derivation of the Slater 2022 line-source plume model that will be applied in later chapters to Upernavik Fjord. The intention is for this chapter to have a stand-alone capacity, to serve as an introductory resource for students interested in understanding the basics of BPT or employing plume models in their own research.

1 The Main Assumptions

The roots of BPT are often traced back to a 1956 paper by Morton, Taylor, and Turner, titled *Turbulent Gravitational Convection from Maintained and Instantaneous Sources* [5]. This paper derives a set of self-similar solutions referred to as the MTT model and makes a key choice about the parameterization of entrainment. The MTT model is based on a few important assumptions, discussed here.

(i) The Entrainment Problem (and Solution)

As a buoyant plume rises, it entrains a large amount of ambient fluid through turbulent mixing processes. This entrainment causes the plume's width to increase, its density to change, and its other properties to be diluted towards the ambient properties. Being able to parameterize the entrainment of ambient fluid into a plume was, and continues to be, an important challenge in the attempt to model them. The small-scale turbulent dynamics are far too complex to model directly, so simpler workarounds have been proposed and tested over time. One of these, proposed in the 1956 MTT paper, is far and away the most widely used thanks to its simplicity and remarkable accuracy when compared to real-life data. The MTT entrainment assumption states that the rate of entrainment into a plume is proportional to the plume's vertical velocity, scaled by a single constant α [5].

While the proportional relationship between entrainment rate and velocity has been tested repeatedly over the decades and has generally proved to be a good description of reality, disagreement persists regarding the exact value of the entrainment constant α [6]. Even among the various papers cited here describing freshwater plumes in the ocean, the values range from 0.036 ([3], [7]) to 0.08 ([8]), to 0.1 ([9], [2]), to 0.13 ([4]) and more.

(ii) Self-similarity

Self-similarity is a property of certain mathematical shapes and functions defined by the lack of a characteristic scale. A self-similar shape is the same at all magnifications: for instance, fractals are by definition exactly self-similar. In an analytical context, self-similar relationships can reduce mathematical complexity. Rising plumes lend themselves well to a self-similar description, as their behavior far from the source is notably scale-independent (the horizontal distribution of velocity, temperature, etc., scaled by the plume width, is effectively identical at each height).

A self-similarity assumption is useful for the MTT entrainment parameterization, as it provides a consistent description of the velocity at the plume's edge [5]. It continues to be a useful property well into the days of recursive computer modeling because of its capacity to reduce the dimensionality of a system of equations. For instance, the model described here is considered constant in the horizontal along-front direction, leaving the vertical dimension z the only independent variable [7].

(iii) The Boussinesq Approximation

A Boussinesq plume is a special case of buoyant plume where the density difference between the plume and the ambient fluid is small. As stated in Jenkins 1991, the density variation is assumed "only to alter the weight of the water and not its mass" ([3]). This seemingly paradoxical statement simply means that mathematical terms that use density to calculate buoyancy are kept (they are the driver of the motion, after all), while the variation in any other terms that depend on density (i.e. momentum) is assumed to be negligible in comparison. This assumption significantly simplifies the mathematical description, and is widely accepted for many plumes in nature, including ours, where $|\rho_{plume} - \rho_{ambient}| << \rho_{ambient}$ [10].

2 The "Full" 1-Dimensional Plume-Melt Model

The mathematical structure presented here is for a line-source plume with the specific application to ice-ocean interactions like the one formulated in Jenkins (1991) [3] and then Jenkins (2011) [7], discussed in Cowton (2015) and Slater (2016) (which derives an analogous model for a point-source plume) [11], [12], and applied by Slater (2022) to a variety of Greenlandic fjord systems [1]. In this last, the line-source model was chosen over a point-source after previous studies ([13], [14]) indicated that the choice of a line with a width on the order of several hundred meters was a more accurate conception of Greenlandic SGD source geometry. This model does not assume a uniform ambient water structure, and can even be forced using a real CTD profile of temperature and salinity. This is a more accurate representation of a Greenlandic fjord, which are typically strongly stratified, but must be solved computationally using recursive ODE solvers (though it remains highly idealized and is not computationally intensive). Simplified versions that can be solved analytically or in 1D are discussed in section 3.

Step 1: Principles of Conservation

Since its genesis with the MTT paper in 1956, the mathematics of buoyant plume theory typically begins with a set of simple laws of conservation, eqns (1a)-(1d). These are, respectively, conservation of volume (1a), momentum (1b), and heat (1c) fluxes. When these equations were applied in the context of ice-ocean interactions, such as in Jenkins (1991) [3], a fourth equation was added for the conservation of salt flux (1d).

$$\frac{\partial Q}{\partial z} = \frac{\partial}{\partial z}(bu) = \alpha u \sin(\theta) + \dot{m} \tag{1a}$$

$$\frac{\partial M}{\partial z} = \frac{\partial}{\partial z}(bu^2) = bg' \sin(\theta) - C_d u^2 \tag{1b}$$

$$\frac{\partial (QT)}{\partial z} = \frac{\partial}{\partial z}(Tbu) = \alpha u \sin(\theta) T_a + \dot{m} T_b + C_d^{1/2} u \Gamma_T (T - T_b) \tag{1c}$$

$$\frac{\partial (QS)}{\partial z} = \frac{\partial}{\partial z}(Sbu) = \alpha u \sin(\theta) S_a + \dot{m} S_b + C_d^{1/2} u \Gamma_S (S - S_b) \tag{1d}$$

In the above equations, b and u are the thickness (the term "width" is used for the other dimension, the width of the source line) and upwards/along-front velocity of the plume in units of

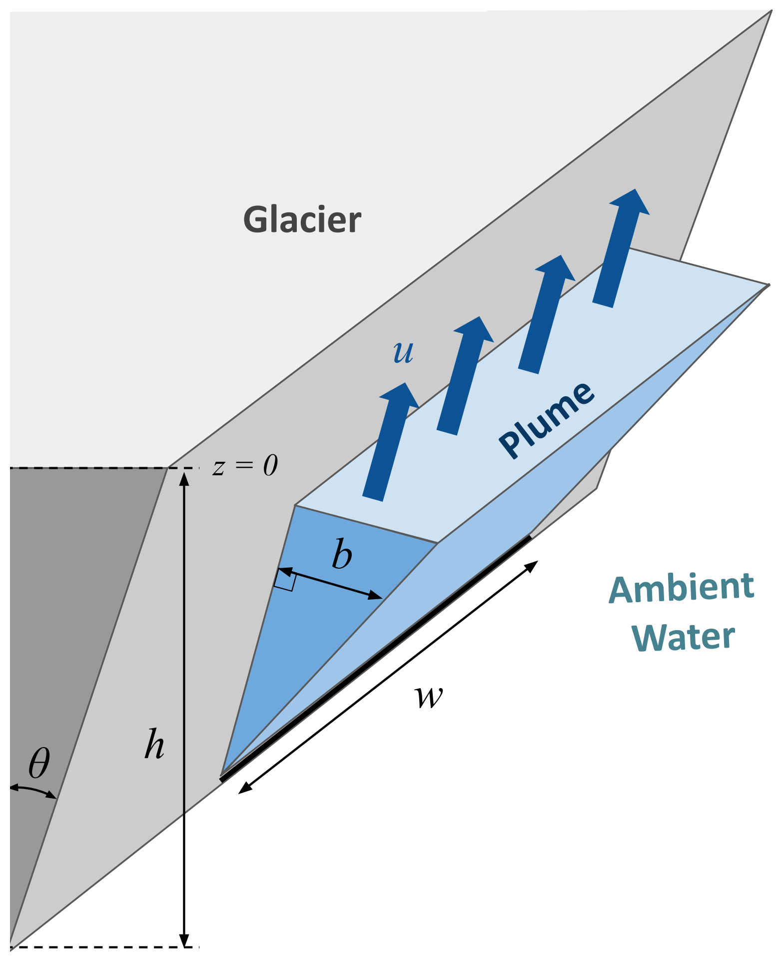

Figure 1: Schematic for a line-source plume rising along an inclined glacier face. The relevant dimensions for the plume model are labeled. The z-dimension is vertical and negative downward with $z = 0$ at the surface of the ambient water. The horizontal distance w is the width of the source line and b is the plume thickness. u is the along-front velocity of the plume and h is the glacier grounding depth, measured from the source depth to the surface of the ambient water.

m and m/s, respectively (Figure 1). Q is the volume flux[1] (units of m^2/s), M is the momentum flux (m^3/s^2), QT is the heat flux where T is the temperature in °C, and QS is the salinity flux where S is the absolute salinity in g/kg.

Equation (1a) describes how the volume flux of the plume (the volume of water that passes through a horizontal rectangle with dimensions 1×b each second) changes with height from the source. This model considers two volume contributions to the plume: entrainment of ambient water (the first term), and meltwater from the glacier front (second term). The entrainment term follows the MTT entrainment assumption discussed in the previous section, where α is the entrainment coefficient (typically taken to be 0.1) and θ is the angle of the glacier front (Figure 1). The melt rate $\dot{m}$ is discussed below.

Equation (1b) describes the conservation of momentum flux[2] of the plume. The two contributors to momentum are buoyancy (first term) and drag (second term). The buoyancy, or upwards gravitational acceleration, is quantified by the reduced gravity $g' = g\frac{(\rho_a - \rho)}{\rho_{ref}}$, where ρ is the density of the plume, ρ_a is the density of the ambient water, and ρ_{ref} is a Boussinesq reference density (for instance, the average density of seawater) [12]. The factor of b is from horizontally integrating g' across the plume thickness, and the $\sin(\theta)$ is the along-front component of the gravitational acceleration. The drag term uses a simple quadratic parameterization with an empirically derived constant drag factor C_d [3].

Equation (1c) describes the conservation of heat flux in the plume, where heat is added or removed by entrainment with ambient water and melting. The first two terms mirror equation (1a): the first is the entrainment rate of ambient water multiplied by the temperature of the ambient water T_a (which may change with height), and the second is the melt rate multiplied by the ice-ocean boundary temperature T_b, which is defined below. The final term of equation (1c) parametrizes the heat lost to the glacier front by turbulent mixing in the boundary layer, parameterized by the heat-transfer coefficient Γ_T, which will also be discussed later.

Equation (1d), which describes conservation of salt flux, has the same form as equation (1c) because the mechanisms of salt and heat transport in the plume are very much related: both are modified by entrainment of ambient water, melting, and turbulent mixing along the ice-ocean boundary.

This system of ODEs contains much of the important information that describes how the plume evolves, but some additional relationships are needed to evaluate them.

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[1] Note from units that this is volume flux per unit width.

[2] Though momentum flux is harder to visualize than volume flux, we can gain an intuitive understanding of it by remembering the kinematic definition *momentum* = *mass* × *velocity*, where mass is calculated by *volume* × *density* ρ. In the plume model we are interested in volume flux rather than volume, but we can consider them the same way: Technically, to calculate momentum flux, we should multiply the volume flux Q by its density and velocity. However, if we recall the Boussinesq approximation described in the previous section, we can ignore changes in ρ when it is not in a buoyancy context, and thus the "constant" ρ divides out of the differential equation. This is why M can be replaced by $Qu = bu^2$ in equation (1b)

Step 2: Ice-Ocean Interactions

This system of ODEs is constrained using a three-equation melt parameterization that relates heat and salt balances along the ice-ocean interface [7]. Thanks to these equations, we will be able to write T_b and S_b in terms of known ambient properties T_a and S_a, physical constants, and plume variables T, S, and Z.

The three-equation melt parameterization is based upon a model developed in 1972 [15] and further developed by a series of other papers [16], [17], [8]. It was first applied to the context of plume modeling in the ice-ocean boundary by Jenkins (1991) [3] and has been employed several times since then [7], [9], [12]. This set of equations attempts to capture the effects of the complex turbulent processes that occur between the rising plume and the ice front—the melting, diffusion, and mixing that impact temperature and salinity fluxes—in a relatively simple way. It is out of the scope of this work to follow the development of this model over time, as it is the result of a long process of making simplifying assumptions, comparing with experiment, changing the assumptions, and so forth. What is useful to us now is the knowledge that these three equations a) close our mathematical system in a simple way and b) are consistent with observed phenomena [17]. The three-equation melt parameterization is shown here and describes the balance of heat (2a) and salt (2b) at the ice-ocean boundary as well as a simplified version of the liquidus relationship (2c) that defines the conditions under which ocean water freezes.

$$\dot{m}L + \dot{m}c_i(T_b - T_i) = c_w C_d^{1/2}\Gamma_T u(T - T_b) \tag{2a}$$

$$\dot{m}(S_b - S_i) = C_d^{1/2}\Gamma_S u(S - S_b) \tag{2b}$$

$$T_b = \lambda_1 S_b + \lambda_2 + \lambda_3 Z \tag{2c}$$

The descriptions and values for the variables in the above equations are shown[3] in tables 1 and 2.

Equation (2a) balances the fluxes of heat into and out of the plume, requiring that the heat used to warm and melt the ice equals the heat transferred from the plume to the ice via turbulent mixing. The left side of the equation is proportional to the melt rate $\dot{m}$[4]; the first term uses the latent heat of fusion L to quantify the heat that went into melting the ice and the second term quantifies the heat used to warm the ice to its melting temperature T_b. (To clarify: the terms in this equation are not strictly measures of heat as they are not in units of Joules, but rather J m kg^{-1}

[3] Γ_T and Γ_S are empirically-derived heat and salt transfer coefficients. When scaled by the square root of the drag coefficient C_d, the resulting quantities are known as the thermal and haline Stanton numbers, respectively, and are dimensionless ratios that describe the efficiency of heat and salt transfer [7].

[4] It may come as a surprise that the melt rate $\dot{m}$ is in units of m/s. As described in Jenkins (1991) [3], this can be thought of as "a thickness of water per unit time;" that is, when integrated over a vertical extent (the glacier front), we obtain a horizontal volume flux (per unit width, like our volume flux from (1a)) of meltwater entering the ocean.

Symbol	Description	Value	Units
α	Entrainment coefficient	0.09	Unitless
C_d	Drag coefficient	2.5×10^{-3}	Unitless
L	Latent heat of fusion for ice	3.35×10^{5}	J kg^{-1}
c_i	Specific heat capacity of ice	2.009×10^{3}	J kg^{-1} K^{-1}
c_w	Specific heat capacity of seawater	3.974×10^{3}	J kg^{-1} K^{-1}
$C_d^{1/2}\Gamma_T$	Thermal Stanton number	1.1×10^{-3}	Unitless
$C_d^{1/2}\Gamma_S$	Haline Stanton number	3.1×10^{-5}	Unitless
λ_1	Seawater freezing point slope (w.r.t salinity)	-5.73×10^{-2}	°C (g/kg)$^{-1}$
λ_2	Seawater freezing point offset	8.32×10^{-2}	°C
λ_3	Seawater freezing point slope (w.r.t depth)	7.61×10^{-4}	°C m^{-1}

Table 1: Physical Constants. Adapted from [7]

s^{-1}). The right-hand side of the equation is related to the heat flux transported from the plume to the ice as a result of turbulent convection, and is parametrized as a function of the temperature difference between the plume and ice and the friction velocity of the rising plume. The friction velocity, related to the shear stress of the plume's vertical motion along the ice front, depends on the dimensionless drag coefficient as $C_d^{1/2}u$ [17].

The same processes are considered from a perspective of salt transport in equation (2b). The left-hand side of the equation is the flux of salt into the boundary layer required to offset the dilution by meltwater in order to maintain the boundary salinity S_b. The right side of the equation states that this salt flux is supplied by turbulent mixing in much the same way as energy is supplied in (2a).

In equation (2c), the boundary temperature T_b is the freezing point of the water along the ice-ocean boundary. This temperature is colder than 0°C, because the nonzero salinity S_b as well as ocean pressure both decrease the freezing point. Equation (2c) is a simple fit of a much more complex relationship, tuned by the empirically determined factors of λ_1, λ_2, and λ_3 for the salinities and pressures experienced in a subglacial environment (equation (2c) fails, notably, at 0 g/kg salinity and 0 m depth: T_b, though small, is not 0°C).

Equations (2a) – (2c) can be used to find an equation for S_b in terms of only known values. To do this, (2a) and (2b) can each be solved for the friction velocity $C_d^{1/2}u$ and set equal to each other, thus eliminating the dependence on u and the melt rate $\dot{m}$. Equation (2c) can then be substituted in for boundary temperature T_b. The resulting relation is a quadratic equation of S_b of the form $K_1S_b^2 + K_2S_b + K_3 = 0$ where coefficients K_1, K_2, and K_3, given in equations (3a) – (3c), depend only on plume variables T, S, and Z.

Symbol	Description	Units	Notes
$\dot{m}$	Melt rate	m/s	Varies with height along interface
S, T	Salinity and temperature in the plume	g/kg, °C	Varies with height and uniform across width
S_i, T_i	Salinity and temperature of the ice	g/kg, °C	Taken as constant with height. Note $S_i = 0$.
S_b, T_b	Salinity and temperature of the ice-ocean interface	g/kg, °C	Varies with height along interface
Z	Depth (distance from surface; negative downward)	m	Varies with height along interface

Table 2: Variables

$$K_1 = -\lambda_1 c_w \Gamma_T + \lambda_1 c_i \Gamma_S \tag{3a}$$

$$K_2 = c_w \Gamma_T (T - \lambda_2 - \lambda_3 Z) + \Gamma_S (c_i(\lambda_2 + \lambda_3 Z - \lambda_1 S - T_i) + L) \tag{3b}$$

$$K_3 = -\Gamma_S S (c_i(\lambda_2 + \lambda_3 Z - T_i) + L) \tag{3c}$$

The positive definite solution of this quadratic relation at a given depth is the boundary salinity S_b. It can be calculated for each height above the plume source by evaluating the quadratic formula $S_b = -K_2 + \frac{\sqrt{(K_2)^2 - 4(K_1)(K_3)}}{2K_1}$ for the corresponding values of T, S, and Z. Once S_b is known, boundary temperature T_b and melt rate $\dot{m}$ at each height can then be calculated using equations (2c) and (2b) respectively.

Step 3: Closing the System

The model defined by the conservation relations and the three-equation melt parametrization is closed by incorporating an equation of state to relate salinity and temperature to density (constraining g') and by assuming a constant drag coefficient C_D [7].

The equation of state of seawater describes the dependence of density on salinity and temperature and is a complex, empirically derived relationship. A linearization of the equation of state like equation (4) (i.e. Jenkins 2011 [7]) can be used to simplify the math, though for models solved computationally the full equation of state is also accessible; for instance, through the use of the `gsw` package in Python.

$$\rho = \rho_{\text{ref}}[1 + \beta_S (S - S_{\text{ref}}) - \beta_T (T - T_{\text{ref}})] \tag{4}$$

Step 4: Initial Conditions

In order to solve the differential equations (1a) – (1d), we must define a set of initial conditions; that is, the properties of the plume (S_0, T_0, u_0, b_0, Q_0) as it exits the source. We know that the SGD is freshwater runoff, so S_0 is 0 g/kg. We assume that the exiting SGD, having spent significant time in contact with the underside of the glacier, is approximately in thermal equilibrium with the ice around it[5]. Both are thus effectively at the pressure freezing point for freshwater as defined in equation (2c): $T_0 = \lambda_2 + \lambda_3 h$ where h is the grounding depth of the glacier (Figure 1). Due to the geometry of the line plume, the initial plume thickness b_0 and vertical velocity u_0 are related to the SGD flux Q_0 as $Q_0 = b_0 u_0$. Both b_0 and u_0 are unknown but can be constrained by considering the dimensionless constant $\Gamma = bg'/\alpha u$, which can be thought of as a ratio of buoyancy to momentum. It has been shown that a plume in unstratified ambient water will quickly approach $\Gamma = 1$, known as a "pure plume," with a balance between buoyancy and momentum [6], [12]. If we assume that $\Gamma_0 = 1$ from the source, this constraint allows us to find expressions for b_0 and u_0 algebraically:

$$b_0 = (\frac{\alpha Q_0^2}{g_0'})^{1/3} \tag{5a}$$

$$u_0 = (\frac{Q_0 g_0'}{\alpha})^{1/3} \tag{5b}$$

Where g_0' is the initial reduced gravity of the plume with density calculated from T_0 and S_0 using either the linearized or full equation of state. Given these initial conditions, equations (1), (2), and (3), along with an equation of state, can be solved computationally for each depth along a plume's vertical extent to create profiles of temperature, salinity, radius, melt rate, and velocity. This model is flexible in that it can be forced with vertically nonuniform temperature and salinity profiles, such as those obtained by CTD measurements close to a glacier front.

3 The One-Point Analytical Model

The analytical model described here, a descendant of the MTT model of buoyant plume theory, is a simplification of the line-source plume model outlined in the sections above (which was first used in [7] and applied to a point-source plume in [12]). However, whereas the "full" model above can be forced with an arbitrary stratification of ambient water properties, the simplified analytical model assumes an unstratified ambient environment. This assumption is based off of the two-layer model of a Greenlandic fjord: warm, salty Atlantic Water (AW) at depth and a thinner layer of cold, fresher, Polar Water (PW) near the surface. The Muilwijk model assumes that the majority of the entrained ambient water in the plume comes from the deeper AW layer, a choice that the authors

[5]Note here that the temperature of a glacier is typically warmer at the bottom, near the bedrock, than it is higher up. This is a result of the increased pressure, of course, but also of geothermal activity occurring in the Earth's crust, which is unusually prevalent in Greenland and Iceland due to a hotspot of tectonic activity in the region [18]. Though interesting, this detail is typically neglected in the context of SGD temperature.

argue is reasonable [2]. Additionally, they ignore certain dynamics in the plume as it rises, choosing only to calculate the fluxes of meltwater, SGD, and entrained ambient water in the plume after it has reached its neutral buoyancy point—thus providing only a single data point, all that was needed to calculate the properties of the resulting water mass. A more mathematical description of the assumptions is outlined here.

Aside: The Case for a Simple Model

Though a "full" model that incorporates detailed physics and a high-resolution output can be appealing, it is not always the most useful or informative. The dependence upon empirically determined constants and simplifications means that the increased resolution can come at the expense of accuracy. In the context of plume modeling, the detail about temperature, salinity, and melt rate throughout the plume's depth may not be necessary for certain applications, and a simpler model might be more a better choice.

It is important to remember that all models are inherently "wrong": they should not be presumed to replicate the actual behavior of a system. A simpler model that makes only a few foundational assumptions can be just as good as—or rather, no worse than–a complex model that makes many small assumptions. At times, a simple model can in fact be even more informative than a detailed one. For instance, a detailed or high-resolution model can give the impression of being more realistic to a reader, just by nature of our innate biases as humans. This can lead to conclusions being made based on observations of details that were misleading or even purely fictitious. Additionally, it is easier to account for a few large, foundational assumptions than for countless small ones. Their relative impact can be assessed and presented concisely, and their physical meaning understood by readers seeking to contextualize a model's results.

One example of a simplified plume model based off of the Slater (2016) "full" model was used by Muilwijk et al. (2022) [2] for application to data from Upernavik Fjord.

Deriving the Analytical Model

The first simplifications were proposed within Slater (2016), with the goal of making a model that was solvable analytically rather than with a computational ODE solver. A more thorough description of these simplifications and the justifications for each can be found in that paper and its supplementary materials [12].

Simplification 1: Ignore the feedback effects of melting. Mathematically, this means dropping the final term on the right hand side of equation (1a) (the volume contribution from glacial melt) and the last two terms each on the right hand side of (1c) and (1d) (which consider the cooling and freshening effects of mixing with meltwater and the related transfers within the boundary region).The authors show that these terms add only a small correction, and can be safely neglected in plumes with significant enough SGD (greater than ≈ 5 m^3/s, as most plumes are) [12].

Simplification 2: Ignore drag feedback along the interface. This allows us to drop the final term on the right hand side of (1b), which is already a simple parameterization using a constant drag factor C_d. The authors justify this choice by demonstrating the small effect this term has on the overall velocity (a decrease of only $\approx 2.5\%$ in uniform stratification) [12].

Simplification 3: Use a linear equation of state. The use of equation (4) rather than the full equation of state for seawater greatly simplifies the analysis while introducing minimal error [7], [12]. Using equation (4) and the definition of reduced gravity $g' = g\frac{(\rho_a - \rho)}{\rho_{ref}}$, we can write a new formula for g' in terms of S and T:

$$g' = g[\beta_S(S_a - S) - \beta_T(T_a - T)] \tag{6}$$

Simplification 4: Assume an unstratified ambient environment. As mentioned previously, the case of the Greenlandic fjord is not unstratified. However, this assumption may be reasonable to describe the beginning of the plume's formation as it rises through the (mostly) uniform AW layer and greatly simplifies the model's solutions [2].

Simplification 5: Assume a vertical calving front. This assumption is not strictly necessary, but simplifies any term proportional to $\sin(\theta)$. It is applied by Muilwijk (2022) because it is likely also an accurate representation of the glacier geometry in Upernavik Fjord[6].

Our new model consists of simplified versions of equations (1a)-(1d) that can now be solved analytically:

$$\frac{\partial Q}{\partial z} = \frac{\partial}{\partial z}(bu) = \alpha u \tag{7a}$$
$$\frac{\partial M}{\partial z} = \frac{\partial}{\partial z}(bu^2) = bg' \tag{7b}$$
$$\frac{\partial (QT)}{\partial z} = \frac{\partial}{\partial z}(Tbu) = \alpha u T_a \tag{7c}$$
$$\frac{\partial (QS)}{\partial z} = \frac{\partial}{\partial z}(Sbu) = \alpha u S_a \tag{7d}$$

To find the equations for b, u, S, and T that satisfy these ODEs, we will first identify some important relationships. For every height in the plume, the total volume flux $Q = bu$ and momen-

[6]Marine-terminating glaciers in Greenland tend to have grounded, approximately vertical calving faces. This is a contrast to the long, floating "ice shelves" more commonly seen in Antarctica [19], which provided the context for many of the BPT predecessors of this model [3], [17], [8].

tum flux $M = bu^2$. We also define a variable called buoyancy flux B where $B = g'Q$ [4]. It can be shown by combining our simplified formula for reduced gravity, equation (6), with (7c) and (7d) that $\frac{\partial(g'Q)}{\partial z} = \frac{\partial B}{\partial z} = 0$. Buoyancy flux is thus constant with height in a uniform water mass[7] and $B = g'Q = g'_0 Q_0$.

Using these relations, we can rewrite (7b) in two ways. First, we get $\frac{\partial M}{\partial z} = \frac{Q_0 g'_0}{u}$ by rewriting b and g' on the right-hand side. Second, using the product rule to expand $\frac{\partial M}{\partial z} = \frac{\partial(Qu)}{\partial z} = u\frac{\partial Q}{\partial z} + Q\frac{\partial u}{\partial z}$, and substituting in eqn (7a), we get $\frac{\partial M}{\partial z} = \alpha u^2 + Q\frac{\partial u}{\partial z}$. These two equivalent equations can be set equal to each other to get:

$$\frac{\partial M}{\partial z} = \frac{Q_0 g'_0}{u} = \alpha u^2 + Q\frac{\partial u}{\partial z} \tag{8}$$

For a line-source plume in an unstratified environment, it can be shown that u must be constant with height, eliminating the last term in equation (8) and allowing us to find an equation for u:

$$u = (\frac{Q_0 g'_0}{\alpha})^{1/3} \tag{9}$$

Now we can evaluate equation (7a): because u is constant, we can integrate to get $Q = \alpha u z + C$. Rewrite this strategically as $Q = \alpha u(z + z_0)$ so that $Q(0) = Q_0$ and $z_0 = \frac{Q_0}{\alpha u}$ is the "source correction" that allows for a finite initial volume flux at $z = 0$. Plugging in our equation for u, we can rewrite the source correction as:

$$z_0 = (\frac{Q_0^2}{\alpha^2 g'_0})^{1/3} \tag{10}$$

Now it is simple to find an equation for b using $b = Q/u$:

$$b = \alpha(z + z_0) \tag{11}$$

Analytical solutions can also be found for T and S of the plume as a function of height[8], but the model used in Muilwijk (2022) is interested only in the relative volume fluxes of entrained water and SGD. Using $Q = bu$, we can write:

$$Q = \alpha(z + z_0)(\frac{Q_0 g'_0}{\alpha})^{1/3} \tag{12a}$$

$$= \alpha z(\frac{Q_0 g'_0}{\alpha})^{1/3} + \alpha(\frac{Q_0^2}{\alpha^2 g'_0})^{1/3}(\frac{Q_0 g'_0}{\alpha})^{1/3} \tag{12b}$$

$$= \alpha z(\frac{Q_0 g'_0}{\alpha})^{1/3} + Q_0 \tag{12c}$$

Because Q_0 is the volume flux of the initial SGD, the first term on the RHS of (12c) is the entrained water volume flux, made up of uniform ambient water. At the surface, $z = h$ and the total flux of

[7]Conceptually, each new entrained parcel has the same density as every other and no new buoyancy is created or destroyed by moving them around. The only source of buoyancy is from Q_0 at the source, and g' decreases exactly proportionally to the increase in Q as ambient water is entrained.

[8]$T = T_0 + (T_a - T_0)(1 - \frac{z_0}{z+z_0})$ and $S = S_a(1 - \frac{z_0}{z+z_0})$. See [12], supplementary material.

entrained water in the plume can be written as[9]:

$$Q_{\text{ent}} = \alpha h (\frac{Q_0 g'_0}{\alpha})^{1/3} \tag{13}$$

We now know the final volume fluxes of the entrained water Q_{ent} and SGD Q_0. The only remaining water mass is submarine meltwater (SMW) from melting of the glacier front.

Slater (2016) [12] motivates a simple approximation for the total melt rate per unit width $\dot{M}$ where $\dot{M} = \int_0^h \dot{m}\, dz$:

$$\dot{M} = Q_{SMW} = A_1[1 + A_2(T_a - T_0)]Q_0^{1/3} h \tag{14}$$

Where $\dot{M}$, in units of m^2/s, is now a volume flux per unit width and can be compared to Q_{ent} and Q_0. Here, A_1 and A_2 are constants calculated by numerically minimizing the difference between (14) and the "full" model; $A_1 = 1.56 \times 10^{-5}$ s$^{-2/3}$ and $A_2 = 0.84$ °C^{-1} [12].

Applying the One-Point Analytical Model

Thanks to equations (13), (14), and the definition $Q_{\text{SGD}} = Q_0$, we can calculate the final volume fluxes of each of the constituent water masses in a subglacial plume. In Muilwijk (2022), the resulting mixture is referred to as glacially modified water (GMW), the properties of which can be calculated from its three components. Because conservative temperature and absolute salinity are conservative tracers, we can write

$$T_{\text{GMW}} = \frac{T_a Q_{\text{ent}} + T_{\text{SMW}} Q_{\text{SMW}} + T_0 Q_{\text{SGD}}}{Q_{\text{ent}} + Q_{\text{SMW}} + Q_{\text{SGD}}} \tag{15a}$$

(15b)

$$S_{\text{GMW}} = \frac{S_a Q_{\text{ent}} + S_{\text{SMW}} Q_{\text{SMW}} + S_0 Q_{\text{SGD}}}{Q_{\text{ent}} + Q_{\text{SMW}} + Q_{\text{SGD}}} \tag{15c}$$

where T_{GMW} and S_{GMW} are effectively the averages of the temperatures and salinities of their three components, where each is weighted by their respective volume flux.

The Polar Water Correction

Muilwijk (2022) also employs one substantial modification of the one-point analytical model to account for the fact that Upernavik Fjord is not, ultimately, unstratified. The authors note that, though the water entrained in the plume *is* primarily from the deeper AW layer, the GMW tends to mix in a large proportion of polar water (PW) from the upper layer shortly *after* the plume

[9]In [2], the variable Q_0 is the *total* volume flux in m^3/s, not the volume flux per unit width as it is used here. This is why, in in Muilwijk (2022), equation (13) is written as $Q_{\text{ent}} = \alpha w h(\frac{Q_0 g'_0}{\alpha w})^{1/3}$, and (14) is written as $Q_{SMW} = A_1[1 + A_2(T_a - T_0)]Q_0^{1/3} h w$ for a known source width w [2].

has reached neutral buoyancy. In this context, PW is defined as the lighter water mass on the shelf bounded by the same isopycnals as the GMW layer. Because this down-fjord mixing is not included the plume model, they account for this addition separately [2]. Using a different method, the authors determined that the GMW from the plume mixes with PW until PW is approximately 41% of the total water mass; they thus calculate a second temperature and salinity for a so-called *Modified*-GMW, itself a weighted average of the plume model output and pure PW properties using the 59%/41% scaling. The *Modified*-GMW properties calculated using this correction appeared to be a good fit for observed data [2].

4 Intrusion Depth

A rising plume will either be sufficiently buoyant that it rises all of the way to the surface of the ocean, or, by entraining ambient water in a stratified fluid, it will lose buoyancy until it ends up at the same density as the ambient water around it. The depth at which this occurs is the neutral buoyancy depth. A plume that reaches neutral buoyancy below the surface will typically overshoot this depth somewhat thanks to its initial velocity, but will eventually slow and sink back down before detaching from the glacier front and flowing horizontally outwards as a buoyancy-driven current [4]. This is how GMW is introduced into the fjord system, and can occur at any depth depending on the plume properties and the ambient stratification. Calculating the neutral buoyancy depth of a plume can help give an idea of the depth at which the GMW will be exported into the fjord, an important variable in broader questions of freshwater export and circulation. Models such as the "full model" estimate the neutral buoyancy depth directly by finding where the density of the plume water is no longer exceeded by the density of the ambient water column[10]. These methods can help constrain the relative importance of different plume variables on GMW intrusion depth [1].

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